

StockWise: AI-Powered Decision System for Inventory Reordering and Waste Reduction

Product Requirement Documentation

1. Product Overview

Small cafes and kiosks primarily manage inventory through manual checking and spreadsheets, which lacks actionable insights. This leads to a persistent struggle in determining which items require immediate restocking, which are tying up working capital, and which purchase quantities will exacerbate waste. Consequently, businesses face avoidable over-ordering, high stockout risks for fast-moving items, and significant time lost reviewing raw data

Decision intelligence and inventory management specifically tailored for small-scale F&B operators, such as cafes and kiosks

StockWise is a web-based decision intelligence system for small cafes and kiosks that need to make daily inventory decisions under waste and cash pressure. The product converts structured inventory records into ranked actions, simulation-based trade-off analysis, and Z.AI GLM-powered explanations that help non-technical operators decide what to restock now, what to buy less of, and what to delay.

The MVP is grounded in a prototype dataset containing 1,000 records over 100 days, covering 10 recurring inventory items and 14 structured fields. This allows the product to demonstrate practical decision support from real inventory behavior without over-claiming advanced forecasting or revenue prediction.

2. Background and Business Objective

For many years, small-scale inventory management has followed a manual, spreadsheet-dependent pathway. Operators have access to stock numbers, but interpreting those signals into "next-best actions" is slow and relies heavily on intuition rather than transparent logic. This manual process is labor-intensive and creates a barrier for non-technical owners who need to visualize the economic trade-offs of their purchasing decisions.

By accelerating the time it takes to identify reorder priorities from hours of manual review to seconds of automated analysis, operators can drastically reduce avoidable losses from perishable stock. Solving this issue allows small businesses to minimize the cash locked in high-value stock and increase overall operational confidence through explainable guidance

StockWise aligns with the modern AI-native development lifecycle by positioning **Z.AI GLM** as the essential intelligence layer that converts raw analytics into human-readable reasoning. It demonstrates practical economic value by showing how AI can help non-technical operators make smarter, data-informed decisions without the complexity of enterprise ERP systems.

3. Product Purpose

StockWise exists to improve daily reorder decisions. Rather than acting as a passive dashboard, the system acts as a decision advisor that combines transparent inventory logic with AI explanation. The product helps operators convert raw inventory signals into understandable next-best actions.

- Reduce time spent reviewing daily reorder priorities.
- Reduce waste-prone over-ordering for perishable or high-risk items.
- Reduce cash locked in high-value stock that does not need immediate replenishment.
- Increase confidence in reorder actions through explainable, item-specific guidance.

3.1 Main Goal of the System

The primary objective of StockWise is to serve as a web-based decision intelligence system that empowers small cafe and kiosk operators to make daily inventory decisions under waste and cash pressure. The system is designed to achieve the following goals:

- **Act as a Decision Advisor:** Rather than functioning as a passive dashboard, the system provides an active partnership that combines transparent inventory logic with AI-driven reasoning.
- **Transform Data into Action:** The system converts structured inventory records and raw signals into ranked, "next-best" actions, helping users identify exactly what to restock, what to buy less of, and what to delay.
- **Optimize Financial and Material Resources:** StockWise aims to reduce waste-prone over-ordering for perishable items and minimize the amount of working capital locked in high-value stock that does not require immediate replenishment.
- **Bridge the Technical Gap:** By utilizing Z.AI GLM to provide plain-language explanations, the system allows non-technical operators to understand complex trade-offs between shortage risks, waste risks, and cash usage without needing to interpret raw data or formulas.
- **Improve Operational Efficiency:** The system seeks to drastically reduce the manual labor and time spent by owners and managers in reviewing spreadsheets and raw inventory logs.

3.2 Target Audience

The primary user is a small cafe or kiosk owner-operator who is personally involved in purchasing and inventory control. This user needs simple, fast, and trustworthy guidance rather than enterprise software complexity.

User	Needs	Pain Points
Small cafe owner / manager	Quick reorder prioritization and cash-aware purchasing decisions	Limited time, spreadsheet dependence, uncertainty about the right order quantity
Kiosk operator	Simple dashboard and action summary	Difficult to interpret stock levels versus actual usage and lead time

Kitchen / operations lead	Clear item-level risk flags	Cannot easily explain why an item is urgent or waste-prone
---------------------------	-----------------------------	--

4. System Functionalities

4.1 Description

The system operates as a **decision intelligence workflow** that moves the user from raw data to informed action. Rather than just being a visual analytics screen, it integrates deterministic inventory metrics with AI-powered reasoning to provide a "decision advisor" experience for non-technical operators. It captures the relationship between stock levels, lead times, and waste risks to generate actionable reorder guidance

4.2 Key Functionalities

Feature	Description	Why It Matters
Decision Dashboard	Shows stock health, low-cover items, waste-cost exposure, and inventory value at risk.	Gives the user a fast operational overview instead of raw rows.
Priority Action Board	Assigns each item a recommended action: RESTOCK_NOW, BUY_LESS, DELAY_PURCHASE, or MONITOR_CLOSELY.	Turns computed metrics into clear next steps.
What-If Reorder Simulator	Lets the user test a proposed order quantity and see coverage, cash outlay, and risk change.	Supports trade-off analysis before placing an order.

AI Explanation Layer	Uses Z.AI GLM to translate complex analytical trade-offs into plain-language rationale and operator-friendly next steps.	Makes the product a true decision-support tool rather than a score table.
Owner-Friendly Manual Entry	Enables users to submit inventory without technical CSV-only fields, aligning manual and file-based inputs under a unified contract	Improves usability for small operators who do not always maintain a prepared CSV file.
Inventory Records Management	Supports the review, editing, and deletion of uploaded or manual records to ensure recommendation accuracy.	Lets users correct mistakes and refresh recommendations without starting the workflow from scratch.
AI powered Chatbot	Extends the GLM layer into interactive guidance, answering grounded questions about current analysis and simulation scenarios.	Extends the GLM layer from explanation-only support into interactive decision guidance.
Supabase Analysis Snapshots	Stores point-in-time analysis runs and item-level recommendation results.	Allows latest and past analysis results to be restored beyond in-memory cache availability.
Persistent Observation History	Stores every validated source row before grouping and ranking.	Preserves historical CSV/manual records and supports trend-aware analysis.

4.3 AI Model and Prompt Design

4.3.1 Model Selection

The system utilizes **Z.AI GLM**. It is chosen because it acts as the essential intelligence layer that converts raw analytics into human-readable reasoning and trade-off explanations that deterministic scores alone cannot provide.

4.3.2 Prompting Strategy

A **bounded structured prompting strategy** is employed. Deterministic logic first computes metrics and recommendations, and the GLM is then asked to explain these pre-calculated facts rather than invent them, which minimizes hallucination risk.

System prompt:

You are StockWise, an inventory decision advisor for small cafe operators. Use only the provided inventory metrics and structured findings. Explain the recommended action clearly and briefly. Do not invent sales, profit, or supplier facts that are not in the input. Return valid JSON only.

User prompt pattern:

```
{
  "task": "Explain the recommended inventory action for a small cafe operator.",
  "item_context": {
    "item_name": "Paneer",
    "category": "Veg",
    "subcategory": "Dairy",
    "unit": "kg",
    "supplier_name": "Supplier A"
  },
  "current_metrics": {
    "current_stock": 12.54,
    "reorder_level": 5.21,
    "daily_usage": 2.24,
    "lead_time": 4,
    "seasonal_factor": 1.30,
    "price_per_unit": 450,
    "waste_percentage": 4.74,
    "days_of_cover": 5.60,
    "inventory_value": 5643.00,
    "estimated_waste_cost": 267.48,
    "lead_time_demand": 11.65,
    "stock_gap_to_lead_demand": 0.89,
```

```

    "reorder_urgency_score": 72,
    "waste_risk_score": 81,
    "recommended_action": "MONITOR_CLOSELY"
  },
  "recent_context": {
    "avg_usage_7d": 2.37,
    "trend_direction": "up"
  },
  "simulation_context": {
    "simulated_order_qty": 3.0,
    "simulated_cash_outlay": 1350.0,
    "simulated_coverage_days": 6.94,
    "simulated_risk_change": "higher waste exposure with limited urgency reduction"
  }
}

```

4.3.3 Context and Input Handling

To manage input size, the system avoids sending the entire 1,000-row dataset. Instead, it sends only the latest item snapshot plus compact historical summaries like 7-day average usage and trend direction. This keeps the AI context window bounded and aligned with competition requirements.

4.3.4 Fallback and Failure behavior

Level 1: If the initial JSON response is malformed, the system retries once with a stricter prompt.

Level 2: If the retry fails, it uses a **deterministic rule-based template** to provide a basic explanation.

Level 3: It ensures deterministic scores and recommended actions remain visible even if the AI explanation is completely unavailable.

Off-topic Control: The system is programmed to refuse off-topic chat requests instead of forwarding unrelated prompts to the model.

5. User Stories and Use Cases

5.1 User Stories

- As a small cafe owner, I want to see which items need attention today so that I do not spend time scanning raw rows.
- As a kiosk operator, I want to understand whether a risky item is urgent because of low stock, long lead time, or high waste exposure.
- As a purchasing decision-maker, I want to simulate a smaller or larger order quantity so that I can balance stock coverage against waste and cash outlay.
- As a non-technical user, I want AI to explain the recommendation in plain language so that I can act confidently without interpreting formulas myself.

5.2 Primary Use Cases

Use Case 1: Inventory Analysis Flow (CSV or Manual)

1. The user uploads an inventory CSV or submits owner-friendly manual inventory entries.
2. The system validates the schema and normalizes the input into a canonical item contract.
3. The system computes derived metrics (e.g., days_of_cover, lead_time_demand) and ranks items on the dashboard.
4. The user reviews the ranked actions (e.g., RESTOCK_NOW, BUY_LESS) on the Action Board.

Use Case 2: What-If Reorder Simulation

1. The user selects a specific item and enters a proposed reorder quantity in the simulator.
2. The backend recomputes the simulated coverage days, cash outlay, and risk impact.
3. **Z.AI GLM** generates a plain-language explanation based on the simulated metrics.
4. The user chooses an action with a clearer understanding of the economic trade-off.

Use Case 3: Inventory Record Management

1. The user reviews editable inventory records after an upload or manual submission.

2. The user updates or deletes specific item records to correct source data mistakes.
3. The system triggers a recommendation refresh for the affected analysis.

Use Case 4: AI chatBot Decision Guidance

1. The user asks a grounded question to the **AI chatBot**(e.g., "Why does the dairy category look risky?").
2. The system provides a response grounded in the current analysis, including supporting points and actionable chat cards for related items.
3. If a simulation was performed, the user can ask the AI chatBot to explain what changed after the scenario.

6. Features Included (Scope Definition)

The following features define the MVP scope of StockWise, ensuring a complete "data-to-decision" workflow for small operators:

- **Decision Dashboard:** A central visual interface that highlights stock health, low-cover items, waste-cost exposure, and inventory value at risk. It uses category and supplier filters to help users inspect specific segments quickly.
- **Priority Action Board:** A ranking system that assigns each item a recommended action—**RESTOCK_NOW**, **BUY_LESS**, **DELAY_PURCHASE**, or **MONITOR_CLOSELY**—based on explainable urgency and waste-risk scores.
- **What-If Reorder Simulator:** A sandbox environment where users can enter a proposed reorder quantity to immediately see recalculated coverage days, cash outlay, and simulated risk impact.
- **GLM Explanation Layer:** A Z.AI GLM-powered service that translates complex metrics and trade-offs into plain-language rationales and operator-friendly next steps.
- **AI Decision Brief:** A dashboard-level summary generated asynchronously that provides a high-level strategy, identifying "Buy Today" priorities and estimated financial impacts.
- **AI chatBot:** A grounded conversational interface that answers specific questions about current analysis results and supports simulation handoff context.
- **Unified Input Contract (CSV & Manual):** Supports both technical CSV uploads (mapping legacy headers) and owner-friendly manual entry flows to accommodate operators without prepared spreadsheets.

- **Inventory Records Management:** A management module that allows users to review, edit, and delete inventory records after submission to correct data before analysis refresh.
- **Persistent Observation History:** Utilizes Supabase to store every validated source row and point-in-time analysis snapshot, allowing for trend-aware analysis and historical data retrieval.
- **Deterministic Fallback System:** A reliability mechanism that ensures the system shows rule-based templates and dashboard scores even if the AI model output is malformed or unavailable.

7. Features Excluded(Scope Control)

To maintain focus on the decision-intelligence MVP, the following features are intentionally excluded from the current scope:

- **Supplier Automation:** The system does not handle automated supplier ordering or the execution of purchase orders.
- **Financial Forecasting:** Revenue and profit forecasting derived from sales transactions are not supported.
- **Enterprise Integration:** Integration with enterprise-level ERP (Enterprise Resource Planning) systems is excluded.
- **Multi-Branch Sync:** Synchronization of stock levels across multiple branches or locations is out of scope.
- **IoT Integration:** Real-time stock sensing via IoT (Internet of Things) devices is not included.
- **Full Mobile App:** While accessible via web, a dedicated mobile application is not supported in this MVP round.
- **Off-Topic AI Chat:** The AI chatBot will deterministically refuse requests that are unrelated to the current inventory analysis.

8. Assumptions and Constraints

8.1. LLM Cost & Context Constraints

- **Context Management:** To keep inference costs manageable and within model limits, the system does not send the raw 1,000-row dataset to the GLM. Instead, it sends only the latest item snapshot and compact 7-day historical summaries.
- **Inference Efficiency:** A key design decision to control token usage is the use of **Bounded Structured Prompting**. Deterministic logic performs all heavy calculations first, so the GLM only processes a small, structured set of pre-calculated metrics.

8.2. Technical Constraints

- **Model Requirement:** The product must use **Z.AI GLM** to remain eligible for competition judging.
- **Platform Limitation:** The MVP is developed as a web-based application; dedicated mobile application support and complex enterprise ERP integrations are currently out of scope.
- **Persistence:** While the system uses Supabase for snapshot storage, the core analysis must remain functional using in-memory cache to ensure reliability during the MVP round.

8.3. Performance Constraints

- **Response Time:** The system is required to analyze a prototype-sized CSV and generate the initial dashboard view in seconds, not minutes.
- **Generation Limits:** To maintain high-quality visuals and rationales, AI explanations are restricted to short, operator-facing rationale rather than long-form analytical reports.

8.4. User Input & Dataset Assumptions

- **Data Representation:** It is assumed that the prototype dataset (1,000 records across 10 items) is sufficiently representative to demonstrate practical decision support for small cafes.
- **Human-in-the-Loop:** The model is unable to generate recommendations without human-provided inventory prompts (CSV or manual entry) and requires human approval before actions are finalized.
- **Data Integrity:** The system assumes that users will utilize the "Review Records" feature to correct source data mistakes, as the AI logic depends on the accuracy of these initial metrics.

9. Risks & Questions Throughout Development

Recommendation Relevance: If the AI recommendations feel too generic, the value of the demo decreases significantly.

- *Mitigation:* This is mitigated by grounding all explanations in specific item metrics and using concrete examples such as Mutton, Paneer, and Eggs.

AI Hallucination: There is a risk that the GLM might invent facts regarding sales, profit, or suppliers that are not present in the input dataset.

- *Mitigation:* The system uses a bounded structured prompting strategy where deterministic logic computes metrics first, and the GLM is only asked to explain those pre-calculated facts. Furthermore, a parser rejects any unsupported claims before they reach the UI.

Model Reliability and Availability: The system depends on the Z.AI GLM API, which may return malformed JSON or become temporarily unavailable.

- *Mitigation:* A multi-level fallback mechanism is implemented: the system retries once with a stricter prompt, falls back to deterministic rule-based templates, and always keeps core dashboard scores visible even if the AI fails.

Data Integrity and Error Handling: The system must handle missing columns, invalid files, or inconsistent data without crashing.

- *Mitigation:* The backend includes modules for schema validation that provide user-friendly feedback and reject invalid files before analysis begins.

Context Management: Sending oversized datasets to the GLM could lead to performance issues or cost constraints.

- *Mitigation:* The system only sends the latest item snapshots and compact historical summaries (like 7-day usage trends) to keep the context window bounded.

Traceability and Trust: Non-technical users may be hesitant to act on AI advice if they cannot understand the underlying logic.

- *Mitigation:* Every recommendation is traceable to visible metrics, ensuring that the product remains a transparent decision-support tool rather than an opaque "black box".